SIEMENS

Opcenter Execution Foundation

Opcenter Execution Foundation Starter Kit 5.1.0

04/2025

Guidelines

This manual contains notes of varying importance that should be read with care; i.e.:

Important:

Highlights key information on handling the product, the product itself or to a particular part of the documentation.

Note: Provides supplementary information regarding handling the product, the product itself or a specific part of the documentation.

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Table of Contents

ID	OpcenterEXFN_StarterKit
Title	Opcenter Execution Foundation Starter Kit 5.1.0
Product Title	Opcenter Execution Foundation
Version Title	
Product Version	
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Revision	
State	Published
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1 Opcenter Execution Foundation Starter Kit Release Notes

1.1 What's New

Opcenter Execution Foundation Starter Kit App version 5.1.0 provides the following new functionalities:

- Upgraded to Mendix Studio Pro 10.18.4
- Updated Marketplace modules version:
  - Administration module 4.1.0
  - Atlas Core 3.16.4
  - Community Commons 10.1.0
  - Data Widgets 2.30.1
  - Nanoflow Commons 5.0.0
  - Web Actions 2.10.0

1.2 How to Setup your Environment to Create a Low Code UI App from Starter Kit

Current Opcenter Execution Foundation Starter Kit 5.1.0is based on Mendix Studio Pro 10.18.4. It is important to initialize solution instances correctly, so that those instances will be compatible with future upgrades to Opcenter EX FN-based releases that will be based on future versions of Mendix Studio Pro.

How to create a Low Code UI App from Starter Kit

- Create a Low Code UI App from Mendix Marketplace.
- Create a Low Code UI App from Mendix Studio Pro.

Create a Low Code UI App from Mendix Marketplace

1. Verify that the version of Mendix Studio Pro you are using is the correct one.
2. Access to Mendix Marketplace and search Opcenter Execution Foundation Starter Kit
3. Click *Start With Template*
4. Enter the name of the App and the description

Create a Low Code UI App from Mendix Studio Pro

1. Verify that the version of Mendix Studio Pro you are using is the correct one
2. Open Mendix Studio Pro
3. Click on *Create New App*
4. Search Opcenter Execution Foundation Starter Kit in the available templates
5. Enter the name of the App and the description

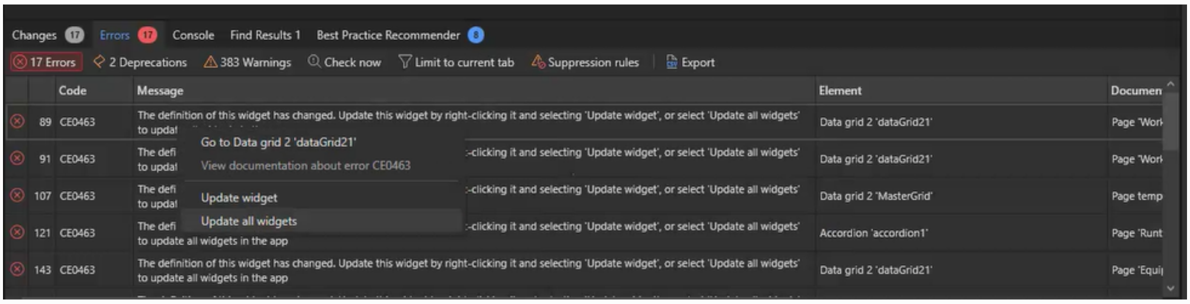
1.3 How to Migrate your Customized Low Code UI App

Workflow

1. Download the latest version of Opcenter Execution Foundation Starter Kit from Support Center or from Mendix Marketplace
2. Verify that the version of Mendix Studio Pro you are using is the correct one.
3. Open the downloaded mpk file in Mendix Studio Pro.
4. If needed, update marketplace modules version to this version.

5. Import in your Low Code UI App the Add-on modules "EXFN_Authentication.mxmodule", "EXFN_ServiceLayer.mxmodule", "EXFN_CrossNavigation.mxmodule" from the folder "STARTER_KIT_APP_DIRECTORY\modules"
6. In Mendix Studio Pro, select the "OpcenterEXFN_DISW_DesignSystem" of the downloaded Opcenter Execution Foundation Starter Kit and click "Export Module package"
7. Import and replace the exported package in your Low Code UI App

> Along the upgrade procedure, you could see errors in Mendix Studio Pro for updated version of Opcenter Execution Foundation Starter Kit widgets. To fast solve the notified issues,
> - select the error
> - right click and *Update All*



1.4 How to Run the Starter Kit Application from Mendix Studio Pro

> The Starter Kit is compatible with Opcenter Execution Foundation v2401 and newer versions.
> If you are upgrading from a a previous version, it is important to refer to the Release Notes for detailed information about the upgrade procedure.
> To customize the Starter Kit and create your own app, refer to the **How to** page in the **Pages** Folder available under the **StarterKit** Module of this App.

When running a Low Code UI Application (e.g. Starter Kit, Complex Manufacturing, Production Terminal etc...) from Mendix Studio Pro, there are some mandatory configurations that must be performed.

Based on where Opcenter Execution EX FN/DS/PR is installed and running, there are two possible configurations:

- Opcenter EX FN/DS/PR and Mendix Studio Pro are installed on different machines. In this scenario, Opcenter EX FN/DS/PR is referred to as a 'remote' machine.

  > In the 'remote' scenario, **HTTPS** configuration is required.

- Opcenter EX FN/DS/PR is installed on same machine as Mendix Studio Pro. In this scenario, Opcenter EX FN/DS/PR is referred to as 'locally' installed.

Prerequisites

To allow Mendix to login with Opcenter EX EX FN/DS/PR when running applications from Mendix Studio Pro, you need to follow these steps:

1. Open the command prompt as an administrator on the Opcenter EX EX FN/DS/PR server.
2. Navigate to the following directory C:\Program Files\Siemens\SimaticIT\Unified\bin

3. Run the following command: **Situaconf.exe -command setcheckredirecturi -enable-check=false**
4. After executing the command, perform an IIS reset to apply the changes.

> - This command in intended to be used only for testing purposes when running locally from Mendix Studio Pro and should not be used in production environments.
> - Make sure you have the necessary administrative rights on the Opcenter EX FN/DS/PR server to execute these commands.

Steps for remote installation

1. Verify that the active configuration is the Remote Configuration (expand the Solution node, select **Settings** and in the **Configurations** tab, make the required configuration active).
2. Edit the remote configuration and go to the **Constants** tab. For all the constants where you see 'MACHINENAME', replace the name with your remote machine host name. See that the following mandatory constants have been set so that the configuration works properly:
   - **EXFN_Authentication.APPBaseUrl**: Represents the Opcenter Execution Foundation Host Name.
   - **EXFN_Authentication.PlantId**: Represents the Opcenter Execution Foundation Plant to which the application should connect.
   - **EXFN_Authentication.UMCBaseUrl**: Opcenter Execution Foundation Host Name or dedicated UMC server
   - **EXFN_Authentication.ClientId**: Represents the Opcenter Execution Foundation Host Name
   - **EXFN_Authentication.AppName**: Represents the Low Code UI Application Name, this along with App Version will be used to retrieve the User Roles in case the application is also deployed on Opcenter Execution Solution Studio
   - **EXFN_Authentication.AppVersion**: Represents the Low Code UI App Version Name (default: unversioned).
3. Save the configuration by clicking **OK**.
4. If the remote Opcenter EX FNFN/DS/PR machine is using a self-signed certificate, from the **App Settings** dialog box, click the **Certificates** tab to import the SSL certificate.
5. Save the configuration and click on the **Run Locally** button present in the menu bar on top of the working environment.

Steps for local installation

1. Verify that the active configuration is the default one (expand the Solution node, select **Settings** and in the **Configurations** tab, make the required configuration active).
2. See that the following mandatory constants have been set so that the configuration works properly:
   - **EXFN_Authentication.APPBaseUrl:** Represents the Opcenter Execution Foundation Host Name **i.e. localhost**.
   - **EXFN_Authentication.PlantId**: Represents the Opcenter Execution Foundation Plant to which the application should connect.
   - **EXFN_Authentication.UMCBaseUrl**: Opcenter Execution Foundation Host Name or dedicated UMC server
   - **EXFN_Authentication.ClientId**: Represents the Opcenter Execution Foundation Host Name
   - **EXFN_Authentication.AppName**: Represents the Low Code UI Application Name, this along with App Version will be used to retrieve the User Roles in case the application is also deployed on Opcenter Execution Solution Studio.
   - **EXFN_Authentication.AppVersion**: Represents the Low Code UI App Version Name (default: unversioned).
3. Click **Edit** and move to the **Constants** tab. If you have an already configured Plant, remember to change the value of the **EXFN_Authentication.PlantId** Constant with the name of your Plant.
4. Click **View App**: the default web browser opens on the application login screen.

Managing the Login behavior through user rights

You can logon to Opcenter EX FN/DS/PR with two types of users, provided they are configured with the required user rights:

- An Opcenter EX FN/DS/PR user, granted with administrative rights and associated to the **SIT_UAF_SYSADMIN** group. In this case the user will directly access the application full contents.
- A Standard Opcenter EX FN/DS/PR user without administrative rights. In this case the user will encounter an application message "No admin user with no Mendix role assigned" unless additional configuration steps are carried out.

This configuration process allows you to login to your application locally with different user Roles.

If you want to logon with Standard users, there are two possible scenarios based on whether the application has been deployed or not on Solution Studio:

- If the Application has never been deployed to Solution Studio:

1. After clicking on **View App Manually**, change the address to **/login3.html** (e.g. http://localhost:8080/login3.html).
2. Login as MendixAdmin (Default user: "**MxAdmin"**, pwd: "**1**").
3. In the **Client Configuration Overview** page, edit your Client Configuration and set the Default Role to **User**.
4. Close the browser and click on **View App** again in Mx Studio Pro.

   After this configuration, when running the app, Mendix will automatically assign the 'User' role to the user who logs in.

- If the Application has already been deployed to Solution Studio at least once:

1. Leverage the integrated role assignment capabilities from Solution Studio. For more information, refer to page *Assigning Low Code UI Application User Roles to Roles* in the *Opcenter Execution Foundation Development and Configuration Guide.*
2. Update the **Constants** under the App Settings as the App Version and Plant Name. They should match to the version deployed on Solution Studio to allow Opcenter Execution FN/DS/PR to correctly assign the user roles assigned in Low Code UI Application User Roles from Role Configuration.

Functional limitation

If your Opcenter EX FN/DS/PR solution is hosted on a remote machine, make sure that it is configured in HTTPS to make the following examples work: Signal, Cross Navigation and Phone Web Layout.

Starter Kit 3.0.0 version is not compatible with previous versions of Opcenter EX FN.

2 Structure of the App

Module	Description
EXFN_Authentication	Add-On module for managing the sign-out while maintaining the association between the terminal and the Equipment.
EXFN_ServiceLayer	Add-On module that makes it possible to use the Opcenter EX FN Service Layer to: • manage errors • recover files • execute reading functions • call commands • execute OData queries directly.
EXFN_CrossNavigation	Add-On module used to manage the navigation between a Low Code App to a Classic App at runtime.
OpcenterEXFN_Core_Delta	Module used to enhance the layout and style provided by the **Atlas_Core** and **DataWidgets** modules so that the graphical characteristics of Siemens-based interfaces are respected.
OpcenterEXFN_DISW_DesignSystem	Module that provides the App with Siemens theme-based interfaces and enriches the **DISW_DesignSystem** module. It also provides: • templates to generate master data screen with a proper layout • the landing page based on MX Navigations settings • the capability to dock user and plant info at bottom of the UI Screen
OpcenterEXFN_MasterData_Connector	Module that provides set of examples to invoke business logic for Unit of Measures.
OpcenterEXFN_MasterData	Module that provides examples for Unit of Measures Managements: • master data screens • master data screens for mobile support
StarterKit	Module that provides *How to* and *Release Notes*

3 Content of the App

Further documentation is available in Opcenter Execution Foundation Starter Kit App.

Use Case/Area	**Opcenter Execution Foundation Starter Kit Page**
Configure the App to be connected to Opcenter Execution Foundation Based Product	Read Me
Troubleshooting	Read Me
How to configure the SWAC Navigation	How To
How to query data, call commands	How To
How to configure the subscriptions	How To
How to configure the landing page	How To
How to use the reading functions with pagination support	How To
How to use the Tree Widget	How To
How to use the Gantt Widget	How To

3.1 Data Archiving Listener Widget

OPC EX FN Data Archiving Listener

Used to change the general behavior of the content inside the widget.

Tab	**Parameter**	**Description**
General	**Behavior**	Choose the behavior of the content inside the widget (Refresh, Hide or Custom behavior).

3.2 Dropdown Widget

OPC EX FN Dropdown

Used to retrieve entities from the database.

The parameters to configure the **OPC EX FN Dropdown** widget are listed in the table below:

Tab	Parameter	Description
General	**Show label**	If set to **True** a Label is shown.
	Label caption	Value displayed as Label.
	Editable	Configuration to make the field editable.
	Is Required	If set to **True** the field is shown as mandatory.
	With Search	If set to **True** the **Search** edit box is displayed.
	Customize	If set to **True** the content shown when an entity is selected can be customized.
DataSource	**Query Datasource**	Data source to retrieve the entities. You can select data from Database, Microflow, Nanoflow, or Association.
	Filter Attribute	Attribute to be displayed in the field when an entity is selected.
	Preselect Value	(Optional) Default entity selected when the page is loaded.
	Attribute to Keep	Attribute set in the field when an entity is selected.
	Placeholder	(Optional) Text displayed in the field before it is activated.
Output Datasource	**Attribute to Write**	Attribute where the selected value (**Attribute to Keep**) is stored.

3.3 File Upload Widget

OPC EX FN File Upload

Used to load a file from the file system, for example loading a text document.

The parameters to configure the **OPC EX FN File Upload** widget are listed in the table below:

<table>
<tr><th>Tab</th><th>Section</th><th>Parameter</th><th>Description</th></tr>
<tr><td rowspan="4">General</td><td rowspan="4">General</td><td>Name</td><td>Name to be assigned to the widget instance you are creating.</td></tr>
<tr><td>Editable</td><td>Setting for rendering the widget field associated editable or not.

Available settings:
Default: The widget field will be editable on the basis of how the Content parameter found in the Datasource section is configured.
Never: The widget field is not editable.
Conditionally: The field associated can be edited on the basis of the following settings:
always: If this is selected, the field is always editable.
based on attribute value: The field is editable only if a condition for a specific attribute is satisfied. The attribute can be either a boolean (true, false) or an enumeration (having a specific value).
Click the Select button, select the Attribute that you want and confirm with OK, then check the Enable options for the values listed.
based on expression: Enter a specific condition either directly or using the Expression wizard button.</td></tr>
<tr><td>Visible</td><td>Setting for rendering the widget field associated visible in the panel.

Click the EDIT button and select one of the following options:
always: If this is selected, the field is always editable.
based on attribute value: The field is editable only if a condition for a specific attribute is satisfied. The attribute can be either a boolean (true, false) or an enumeration (having a specific value).
Click the Select button, select the Attribute that you want and confirm with OK, then check the Enable options for the values listed.
based on expression: Enter a specific condition either directly or using the Expression wizard button.
By selecting Show OPC Ex FN Upload for selected roles, you can choose the role(s) for which the field is visible.</td></tr>
<tr><td>Acceptable Types</td><td>Extension types that are acceptable for the file.</td></tr>
</table>

Tab	Section	Parameter	Description
	Label	**Show Label**	If set to **Yes**, it's mandatory to insert the **Label Caption.**
		Label Caption	Caption label to be shown in the panel, over the widget button. Clicking the **Edit** button, it's possible to type the label name in the **Template** field. It's also possible to insert the label name as a parameter, configuring it in the **Parameters** section.
	Datasource	**Content**	Datasource content attribute. Clicking the **Select** button, a list of available Mendix Data Model attributes is displayed, choose the one related to the source entity.
	Validation	**Is Required**	If set to **YES**, it's mandatory to insert the file entity in the widget.
		Required Message	The text message which has to be shown when the mandatory content is missing. Clicking the **Edit** button, it's possible to type the text message in the **Template** field. It's also possible to insert the message as a parameter, configuring it in the **Parameters** section.
	File Metadata	**File Name**	File name metadata attribute which is shown in the widget. Clicking the **Select** button, a list of available Mendix Data Model attributes is displayed, choose the one related to the source entity.
		File Size	Fle size metadata attribute which is shown in the widget. Clicking the **Select** button, a list of available Mendix Data Model attributes is displayed, choose the one related to the source entity.
		File Type	File type metadata attribute (i.e. the file extension) which is shown in the widget. Clicking the **Select** button, a list of available Mendix Data Model attributes is displayed, choose the one related to the source entity.
	Events	**On Document Uploaded**	Setting to define the event to be triggered when uploading the file entity. Clicking the arrow icon on the right, a list of available events is displayed.
Advanced	**Advanced**	**Custom Upload Button**	Alternative custom upload button configuration, the default is **None**. This option can be used to show an icon instead of the edit box. Clicking the **Select** button, a list of available Mendix Data Model icons is displayed.

Tab	Section	Parameter	Description
Localiz ation	**Localization**	**Upload Button Label**	Label name to be shown in the upload file button. Clicking the **Edit** button, it's possible to type the label name in the **Template** field. It's also possible to insert the label name as a parameter, configuring it in the **Parameters** section.
		Upload Placeholder	Placeholder text which has to be displayed in the widget when no file is selected. Clicking the **Edit** button, it's possible to type the text in the **Template** field. It's also possible to insert the text as a parameter, configuring it in the **Parameters** section.
		File Clear Button Tooltip	Tooltip for the remove icon, which has to be displayed when the file is removed in the widget using the **X** icon. Clicking the **Edit** button, it's possible to type the text message in the **Template** field. It's also possible to insert the text message as a parameter, configuring it in the **Parameters** section.

3.4 Gantt Widget

OPC EX FN Gantt

Used to track tasks over time. It displays tasks as horizontal bars along a timeline, allowing users to see task durations, start and end dates, dependencies, and the overall project progress at a glance.

Tab	Section	Parameter	Description
Views	**Visible Time Interval**	**View start Date**	Sets the start date for the Gantt view.
		View end Date	Sets the end date for the Gantt view.
	Views	**Use custom views**	Enables custom views for different date ranges.
		Selected View	Specifies the initial view option.
Columns	**Left grid settings**	**Sort for name column**	Sets sorting order for the name column.
		Initial left grid width	Defines the starting width of the left grid.

Tab	Section	Parameter	Description
		On task column click	Action triggered on column click.
Tasks	**General**	**Task Datasource**	Data source providing tasks.
		Task ID	Unique identifier for each task.
		Task Datasource selected	Data source for selected tasks.
	Context mapping for changed values	**Task last ID**	Stores the last task ID.
		Task last start date	Stores the last task start date.
		Task last end date	Stores the last task end date.
	Task planned	**Task start date**	Planned start date of the task.
		Task end date	Planned end date of the task.
		Task estimated duration	Planned duration of the task.
		Task title planned	Planned title for the task.
		Task editable	Makes the task editable.
		Task planned CSS Class	CSS class for styling planned tasks.
		Task planned extra Style	Additional style for planned tasks.
	Task Actual	**Use actual tasks**	Enables tracking of actual task data.

Tab	Section	Parameter	Description
Gantt	**Features**	**Default command bar**	Enables the default command bar.
		Current time line	Shows the current time line on the Gantt chart.
		Auto switch views	Automatically switches views as needed.
	Settings	**Zoom factor**	Controls the zoom level of the Gantt chart.
	Styling	**Column lines**	Shows lines between columns.
		Striped rows	Displays rows with alternating colors.
		Row height	Sets the height of each row.
		Bar margin	Sets the space between task bars.
	Tooltip	**Tooltip type**	Defines the type of tooltip to display.
	Scale to content	**Scale to content**	Scales the view to it's content when new tasks are loaded or the view changes.
		Scale to content	Allows you to directly scale to contents or to do it the next time a datasource is refreshed.
Localization	**Date formats**	**Locale**	Sets the locale for date and time display.
		Date time format	Format for displaying date and time.
		Date format	Format for displaying dates.
		Hours format	Format for displaying hours.

Tab	Section	Parameter	Description
	Translations	**First/name column header**	Label for the name column header.
		Start time	Label for task start time.
		End Time	Label for task end time.
		No Time Ranges	Message shown when no time ranges are available.
		Skills	Label for skills information.
		Users	Label for user information.
		Estimated Time	Label for estimated task time.
		Actual Time	Label for actual task time.
		no end	Text displayed when no end time is set.
		No records to display	Message shown when no tasks are available.
		to	Text for the range separator (e.g., start to end).
	View label translations	**Skills**	Label for skills in the view.
		Hour	Label for hourly view.
		Day	Label for daily view.
		Week	Label for weekly view.
		Year	Label for yearly view.
		Time Axis	Label for the time axis.

3.5 Graph Widget

OPC EX FN Graph

Used to graphically display dependencies between production items, for example between Work Order Operations.

The parameters to configure the **OPC EX FN Graph** widget are listed in the table below:

Tab	Section	Parameter	Description
Graph	**General**	**Parent node Id**	If a graph of nested nodes needs to be rendered a parent node id should be supplied. This can then be used in the edges and nodes data sources to retrieve the correct data. The widget will also change this attribute when double-clicking a node.
		Zoom to node	When a change to this attribute is detected the widget will zoom to the node ID supplied in the attribute. After this the attribute will be cleared.
		Zoom nodes	Defines how many nodes around the target node (the node to zoom in on) should be visible on both the left and right side.
	Overrides/advanced	**Graph min height**	Minimum height of the graph.
		Node width	Width of the node.
		Node height	Height of the node.
		Node spacing x	Horizontal space between nodes.
		Node spacing y	Vertical space between nodes.
Nodes	**Nodes**	**Datasource**	Data source to retrieve the nodes for the graph. You can select data from Database, Association, Microflow, Nanoflow, or Xpath.
		Node ID	Property of the data-source item to be used as unique identifier for the node.
		Has children	Property of the data-source item to indicate if certain node has children.
		CSS Class	Option to apply a CSS class to a specific node. Can be achieved by adding an expression based on the item values (attributes).

Tab	Section	Parameter	Description
	Actions	**On click action**	Triggers an action (for example a nanoflow, microflow, or a Show page action) when the end-user clicks on the node.
		On double-click	Triggers an action (for example a nanoflow, microflow, or a Show page action) when the end-user double clicks on the node.
Edges	**Edges**	**Datasource**	Data source to retrieve the edges for the graph. You can select data from Database, Association, Microflow, Nanoflow, or Xpath.
		From ID	Property of the datasource item to indicate the the node ID representing the From end point.
		To ID	Property of the datasource item to indicate the To end point.
		Edge Label	Property of the datasource item to specify a label for the edge.
		CSS Class	Option to apply a CSS class to a specific edge. Can be achieved by adding an expression based on the item values (attributes).
Breadcrumbs	**Data**	**Datasource**	Data source to retrieve the breadcrumb for the graph. You can select data from Database, Association, Microflow, Nanoflow, or Xpath.
		Node ID	Property of the datasource item to indicate the Node Id for a specific level.
		Label	Property of the datasource item to indicate the label for a specific level.

3.6 Lazy Load Widget

OPC EX FN Lazy Load

Used to load solely those parts of the UI that are currently visible inside the screen: this is achieved by setting a minimum height differing from 0. If a portion of the UI falls outside the frame of the monitor, it will not be loaded.

By setting an offset, those parts of the UI that are located immediately below what is currently displayed on screen will be "pre-loaded": once you start using the scrollbar and the parts of the UI become visible, loading is activated.

It is possible to have a progress bar, spinning wheel or similar control (or text) displayed while loading is in progress.

The parameters to configure the **OPC EX FN Lazy Load** widget are to be set in the property grid of Mendix Studio Pro and are listed in the table below:

Tab	Parameter	Description
Configuration	**Min height**	Minimum height for loading only those UI elements currently displayed on screen.
	Margin for lazy loading	Offset (expressed in pixels) that triggers the "pre-loading" of those UI elements found immediately below what is currently displayed on screen

3.7 Signal Subscription Manager Widget

OPC EX FN Signal Subscription Manager

Used to subscribe the page to one or more Signals coming from Opcenter Execution Foundation Apps. For each Signal, it is possible to associate a specific action that must be performed on the UI upon receiving the Signal's message: for example, refreshing data in the UI.

The parameters to configure the OPC EX FN Signal Subscription Manager widget are listed in the table below:

Tab	Parameter	Description
Configuration	**Access Token**	The authentication token that must be set to permit access to Opcenter EX FN in order to subscribe to Signals.
	EX FN url	The URL of the Opcenter EX FN host that has to be retrieved from the client configuration.
	Signal subscriptions	List of all those Signals to which the user wants to subscribe. For each Signal, the following parameters can be set: • **Signal Name**: Name of the Signal. • **App Name**: Name of the Opcenter EX FNApp that publishes the Signal. • **Unsubscribe Signal**: If set to True, the Signal will be unsubscribed if it was previously subscribed. • **Subscription Filter**: (Optional) String used to filter messages. • **Action**: The action that must be performed when the Signal is received. • **Event Output**: Entity in which the message received from the Signal must be stored.

Tab	Parameter	Description
Advanced	**Enable Heartbeat**	Enables/disables heartbeat.
	Maximum number of Attempts	The maximum allowed number of reconnection attempts.
	Retry after Delay	The time wait (expressed in ms) before re-attempting the request.
	Unauthorized callback	Action to be executed when the user is unauthorized.

3.8 Timer Widget

OPC EX FN Timer

Tab	Parameter	Description
General	**Start/Stop Timer**	Boolean attribute to check if the timer is enabled or not.
	Timer Duration (ms)	The time needed (expressed in ms) before completing its action.
	On Timer Elapsed	The action that will be executed on timer clock.

3.9 Tree Widget

OPC EX FN Tree

Used to graphically display hierarchical data in a tree-like structure, allowing user to navigate through parent-child relationships visually.

Tab	Section	Parameter	Description
Nodes	**Required**	**Data Source**	Defines the data collection for tree nodes.
		Node ID	Unique identifier for each node.
		Node parent ID	Unique identifier for the parent node.
		Node CSS Class	CSS class for node styling.
	Overrides (Optional)	**Node Order**	Sets the display order of nodes.
		Node initially expanded	Boolean value to choose if the tree is expanded or not.
		Node disable checkbox	Disables the node's checkbox.
		Node disabled	Disables the node.
		Node selectable	Allows node selection.
		Node enable checkbox	Enables a checkbox for the node.
	Events (Optional)	**On expand**	Action triggered when node expands.
		On right click	Action triggered on right-click.
General	**Features**	**Node selection**	Enables node selection.
		Drag and drop	Allows dragging and dropping nodes.
		Lazy loading	Loads nodes only when needed based on scroll.
		Custom Icons	Adds custom icons to nodes.
	UI	**Empty list message**	Message shown when no nodes are present.

Tab	Section	Parameter	Description
	Dynamic widget settings **(Optional)**	**Show line**	Displays lines between nodes.
		Show icon	Displays icons for nodes.
		Default expand all	Expands all nodes by default.
		Height	Sets the widget's height.
	Translations	**Loading...**	Text shown while loading nodes.

3.10 Visual Inspection Widget

OPC EX FN Visual Inspection

Used to capture the visual inspection details or the failure details, such as: failure id, color, existing coordinates, rows and columns contained in the grid, new coordinates. These details are required when performing quality inspection of visual characteristics during the execution of a Work Order Operation or Step.

The parameters to configure the **OPC EX FN Visual Inspection** widget are listed in the table below:

Tab	Section	Parameter	Description
General	**General**	**Data Source**	The data source for the visual inspection details such as: failure id, color, coordinates etc.
	Dots	**X coordinates**	The X coordinate value of the dot marked on the visual detected failure.
		Y coordinates	The Y coordinate value of the dot marked on the visual detected failure.
		Color	The color information of the visual detected failure.
		Id	The identifier of the visual detected failure.
	Grid Configuration	**Rows**	The number of rows contained in the grid.
		Columns	The number of columns contained in the grid.

Tab	Section	Parameter	Description
		Image	The actual image under consideration for visual inspection.
		Dot Radius	The radius of the dot marked on the image for visual inspection.
	New dot	**X coordinates**	The new X coordinate value of the dot marked on the visual detected failure.
		Y coordinates	The new Y coordinate value of the dot marked on the visual detected failure.